

Dokumentasi Struktur Proyek Absensi TAAT

▢ ROOT — File Utama

File	Fungsi
artisan	CLI Laravel — menjalankan migrate, seed, dll
composer.json	Daftar dependency PHP (Laravel, Spatie, DomPDF, dll)
package.json	Daftar dependency frontend (Tailwind, Vite, Alpine.js)
vite.config.js	Bundler frontend (menggantikan Mix)
tailwind.config.js	Konfigurasi Tailwind CSS
postcss.config.js	Processor CSS
.env.example	Template environment variable
.editorconfig	Aturan coding style standar
phpunit.xml	Konfigurasi testing PHPUnit

▢ app/ — Jantung Aplikasi

app/Http/Controllers/

File	Fungsi
Controller.php	Base class semua controller
DashboardController.php	Redirect user ke dashboard sesuai role
ProfileController.php	Edit profil, ganti password, upload foto

app/Http/Controllers/Auth/ — Login & Registrasi

File	Fungsi
AuthenticatedSessionController.php	Proses login
RegisteredUserController.php	Register user baru (sudah dinonaktifkan)
PasswordController.php	Ganti password (setelah login)
PasswordResetLinkController.php	Reset password → kirim link reset
NewPasswordController.php	Reset password via token
VerifyEmailController.php	Verifikasi email

File	Fungsi
ConfirmablePasswordController.php	Konfirmasi password sebelum aksi sensitif
EmailVerificationNotificationController.php	Konfirmasi email verifikasi
EmailVerificationPromptController.php	Halaman pengingat verifikasi email

app/Http/Controllers/Admin/ — Panel Super Admin

Controller	Fungsi
AdminDashboardController.php	Dashboard admin dengan statistik dan grafik
StudentController.php	CRUD data siswa
TeacherController.php	CRUD data guru
SubjectController.php	CRUD mata pelajaran
SchoolClassController.php	CRUD kelas
MajorController.php	CRUD jurusan (RPL, AKL, BDP, dll)
ParentController.php	CRUD orang tua/wali
ScheduleController.php	CRUD jadwal pelajaran
SemesterController.php	CRUD semester
AcademicYearController.php	CRUD tahun ajaran
AcademicYearTransitionController.php	Proses naik kelas (X→XI, XI→XII, XII→Lulus)
AttendanceGateController.php	Absensi gerbang — scan QR/barcode, rekap, export
ImportController.php	Import data dari Excel
ReportController.php	Laporan absensi (PDF/Excel)
SettingController.php	Pengaturan aplikasi
AccountManagementController.php	Manajemen semua akun + reset password massal
SuperAdminController.php	Kelola akun super admin lain
GuruPiketAccountController.php	Kelola akun guru piket
ActivityLogController.php	Log aktivitas semua user
HolidayController.php	Hari libur nasional (sinkronisasi API)
SchoolHolidayController.php	Libur khusus sekolah

app/Http/Controllers/Student/

File	Fungsi
StudentPortalController.php	Dashboard siswa, QR code absen, jadwal, riwayat kehadiran

app/Http/Controllers/Teacher/

File	Fungsi
TeacherPortalController.php	Dashboard guru, absensi mapel, rekap

app/Http/Controllers/Parent/

File	Fungsi
ParentPortalController.php	Dashboard orang tua, pantau anak, rekap harian/bulanan

app/Http/Controllers/GuruPiket/

File	Fungsi
GuruPiketController.php	Scan absen di gerbang (khusus jam piket)

app/Http/Middleware/

Middleware	Fungsi
EnsureActiveUser.php	Cek status aktif user — jika nonaktif, logout paksa
EnsureGuruPiketSetup.php	Pastikan guru piket mengisi identitas sebelum scan

app/Http/Requests/ — Form Validation

File	Validasi Untuk
LoginRequest.php	Login

File	Validasi Untuk
StudentRequest.php	Create/update siswa
TeacherRequest.php	Create/update guru
ParentRequest.php	Create/update orang tua
SubjectRequest.php	Mata pelajaran
SchoolClassRequest.php	Kelas
MajorRequest.php	Jurusan
ScheduleRequest.php	Jadwal
SemesterRequest.php	Semester
AcademicYearRequest.php	Tahun ajaran
HolidayRequest.php	Hari libur
ProfileUpdateRequest.php	Update profil

app/Models/ — Database ORM (Eloquent)

Model	Tabel	Fungsi
User.php	users	User login multi-role (Spatie Permission)
Student.php	students	Data siswa dengan relasi user + parent
Teacher.php	teachers	Data guru
StudentParent.php	parents	Data orang tua/wali
SchoolClass.php	classes	Kelas (X RPL 1, dll)
Major.php	majors	Jurusan (RPL, AKL, BDP, dll)
Subject.php	subjects	Mata pelajaran
Schedule.php	schedules	Jadwal pelajaran
AttendanceGate.php	attendance_gates	Absensi gerbang (tap in)
AttendanceSubject.php	attendance_subjects	Absensi mata pelajaran (di kelas)
AttendanceSubjectDetail.php	attendance_subject_details	Detail absensi per siswa per mapel
AcademicYear.php	academic_years	Tahun ajaran
Semester.php	semesters	Semester ganjil/genap
QrToken.php	qr_tokens	Token QR (expired 30 detik)
ActivityLog.php	activity_logs	Log aktivitas user
Setting.php	settings	Pengaturan sistem
Holiday.php	holidays	Libur nasional
SchoolHoliday.php	school_holidays	Libur khusus sekolah

Model	Tabel	Fungsi
Device.php	devices	Perangkat terdaftar
PetugasPiket.php	petugas_piket	Sesi piket guru
AttendanceRule.php	attendance_rules	Aturan absensi (batas telat)
Notification.php	notifications	Notifikasi sistem
WhatsappTemplate.php	whatsapp_templates	Template pesan WhatsApp
Import.php	imports	Riwayat import Excel
ClassStudentHistory.php	class_students_history	Riwayat perpindahan kelas siswa
SchoolProfile.php	school_profiles	Profil sekolah (nama, npsn, logo)

app/Policies/ — Otorisasi

Policy	Fungsi
StudentParentPolicy.php	Cek apakah ortu berhak lihat data anaknya
StudentPolicy.php	Authorisasi data siswa (super_admin only)
TeacherPolicy.php	Authorisasi data guru (super_admin only)
SchedulePolicy.php	Authorisasi jadwal (guru lihat jadwal sendiri, siswa lihat kelasnya)

app/Services/ — Business Logic

Service	Fungsi
StudentService.php	Query + logic siswa
TeacherService.php	Query + logic guru
ParentService.php	Query + logic orang tua
ScheduleService.php	Query jadwal
SchoolClassService.php	Query kelas
MajorService.php	Query jurusan
SubjectService.php	Query mapel
SemesterService.php	Query semester
AcademicYearService.php	Logic tahun ajaran

Service	Fungsi
AcademicYearTransitionService.php	Proses naik kelas
AttendanceGateService.php	Logic absensi gerbang
AttendanceSubjectService.php	Logic absensi mapel
HolidayService.php	Logic hari libur
ImportService.php	Parse + validasi + import Excel
PasswordGeneratorService.php	Generate password default
QrTokenService.php	Generate + validasi QR token
ActivityLogService.php	Catat log aktivitas
WhatsAppService.php	Kirim notifikasi WhatsApp
LiburApiService.php	Sinkronisasi libur nasional dari API

app/Jobs/

File	Fungsi
SendWhatsAppNotificationJob.php	Queue kirim WhatsApp (biar tidak blocking)

app/Console/Commands/

Command	Fungsi
CleanupExpiredQrTokens.php	Hapus QR token yang sudah expired

app/Exports/

File	Fungsi
AttendanceGateExport.php	Export absensi gerbang ke Excel
AttendanceSubjectExport.php	Export absensi mapel ke Excel
ParentExport.php	Export data orang tua ke Excel
ParentReferenceExport.php	Export referensi orang tua
ImportErrorReportExport.php	Export laporan error import

Templates/ — Template Excel untuk import: - StudentImportTemplate.php - TeacherImportTemplate.php - ClassImportTemplate.php - ScheduleImportTemplate.php - StudentImportWithParentsExport.php - ParentReferenceSheet.php - StudentImportSheet.php

□ **config/ — Konfigurasi Aplikasi**

File	Fungsi
app.php	Konfigurasi umum Laravel
auth.php	Setting authentication
database.php	Koneksi database
session.php	Session (secure, http_only, encrypt)
permission.php	Konfigurasi Spatie Permission
mail.php	Setting email
queue.php	Queue untuk WA notification
filesystems.php	Storage local/public
services.php	API keys layanan eksternal
absensi.php	Config custom aplikasi (QR ttl, jam masuk)
whatsapp.php	Config WhatsApp gateway
logging.php	Setting log
cache.php	Cache driver
laravel-webp.php	Webp image converter

□ **database/**

Folder	Fungsi
migrations/	Riwayat struktur database (42 file)
seeders/	Data dummy/awal untuk testing
factories/UserFactory.php	Factory data user untuk testing

□ **routes/ — Endpoint URL**

File	Fungsi
web.php	Semua route web (admin, guru, siswa, parent, piket)
auth.php	Route auth (login, register, forgot password)
console.php	Command artisan custom

resources/views/ — Tampilan (Blade)

Folder	Halaman
auth/	Login, register, forgot password, reset password, verify email
layouts/	Layout utama, sidebar, navbar
layouts/partials/	Sidebar, navbar, mobile drawer, mobile bottom nav
components/	Komponen reusable (button, input, modal, dropdown)
admin/	Semua halaman CRUD admin + dashboard
admin/academic-years/transition/	Proses naik kelas
admin/attendance/	Scan, manual, today, export PDF
admin/reports/	Laporan absensi PDF
admin/imports/	Import Excel
student/	Dashboard siswa, QR code, jadwal, riwayat
teacher/	Absensi mapel, jadwal, rekap
parent/	Dashboard orang tua, rekap harian/bulanan
guru-piket/	Scan, rekap, setup piket
profile/	Edit profil, ganti password, upload foto
profile/partials/	Form update profil

Role & Alur Aplikasi

Role Pengguna

1. **Super Admin** — Mengelola seluruh data master (siswa, guru, kelas, jurusan, jadwal, import, laporan)
2. **Guru** — Mengisi absensi mata pelajaran, melihat jadwal mengajar
3. **Siswa** — Melihat QR code untuk absen, jadwal, riwayat kehadiran
4. **Parent (Orang Tua)** — Memantau kehadiran anak, rekap harian/bulanan
5. **Guru Piket** — Scan absensi di gerbang sekolah (sesi piket)

Alur Absensi

Gerbang: Siswa scan QR → Guru Piket scan barcode → Tercatat di attendance_gates **Mapel:** Guru buka absen di kelas → Siswa dicatat → Masuk attendance_subjects

Fitur Unggulan

- ☐ Multi-role dengan Spatie Permission
- ☐ QR Token (expired 30 detik, one-time use)
- ☐ Barcode scan via NIS
- ☐ Naik kelas otomatis (X→XI→XII→Lulus)
- ☐ Import data massal dari Excel
- ☐ Export laporan PDF & Excel
- ☐ Notifikasi WhatsApp (via Fonnte/queue)
- ☐ Activity log semua aksi
- ☐ Session terenkripsi
- ☐ Sinkronisasi hari libur nasional dari API
- ☐ Soft Delete + restore data